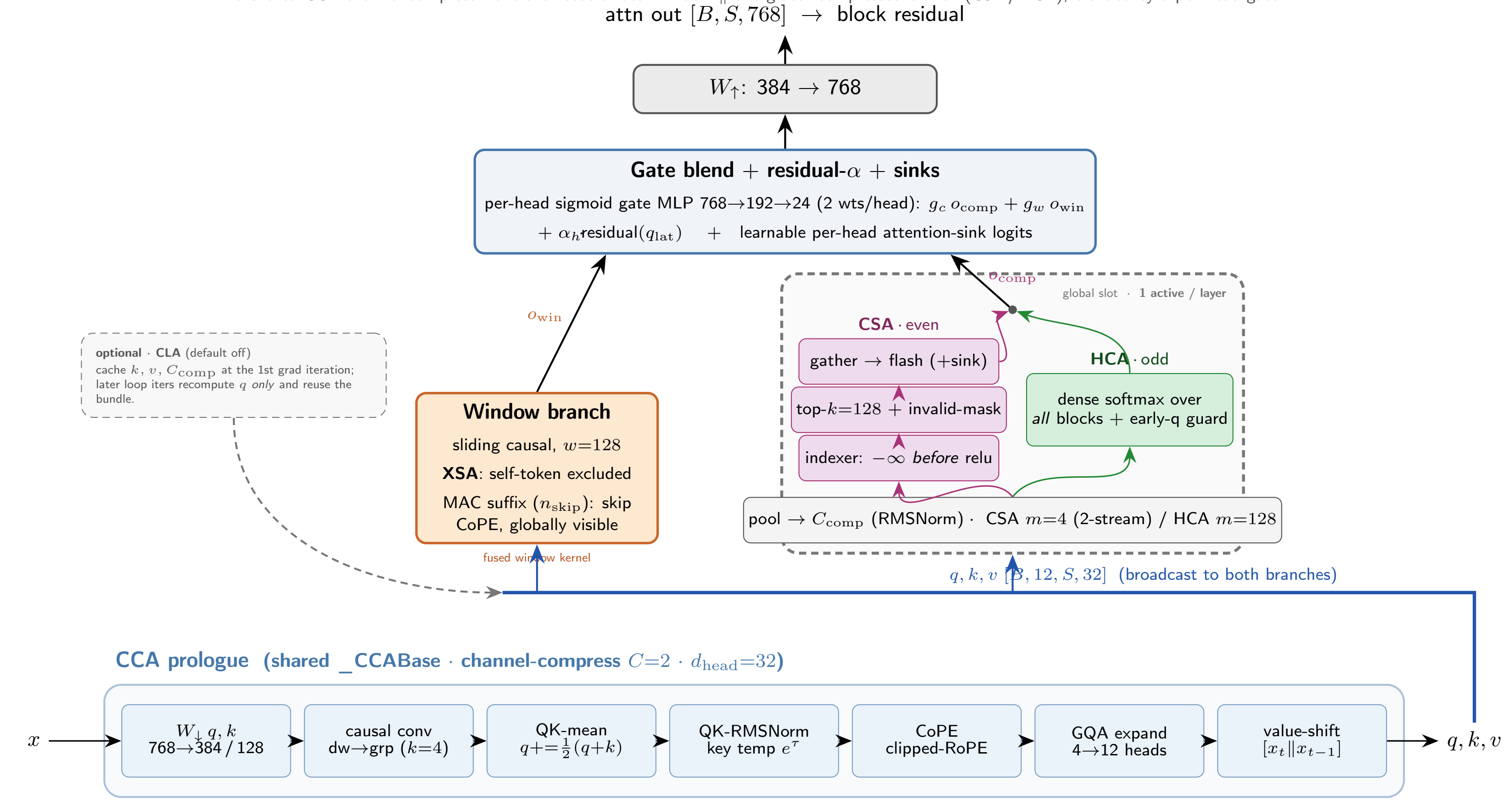
MORPH Attention — Triple-Axis Compression

a shared CCA channel-compress front-end feeds a local window || one global-compressed branch (CSA / HCA), blended by a per-head gate



Four compression axes (faithful to `morph/model/attention.py`): channel CCA $E \rightarrow E/2$ select CSA top-128 of $S/4$ blocks dense HCA all $S/128$ blocks local window w=128.

CSA/HCA alternate on the *global* layer index (even→CSA, odd→HCA), baked at `__init__` — a looped core layer keeps its variant on every iteration (*not* iteration-conditioned). Causal-leak fix: indexer masks future blocks to $-\infty$ *before* relu, *and* re-checks validity after gather.

Combine: per-head sigmoid gate → 2 weights (compressed, window); residual-α adds the down-projected q_{lat} back; attention sinks = a learnable per-head logit on the compressed softmax. Fused Triton: CCA prologue, conv, window, CSA, HCA (each fwd/grad-exact). **Notation:** H=12 q-heads, H_{kv}=4 kv (GQA ×3), C=2, d_{head}=32, $n_{\mathrm{blk}}=S/m$. Color secondary — every box is text-labeled.